PHASE 6 • PROJECT TESTING

✓ VERIFIED SPECIFICATION

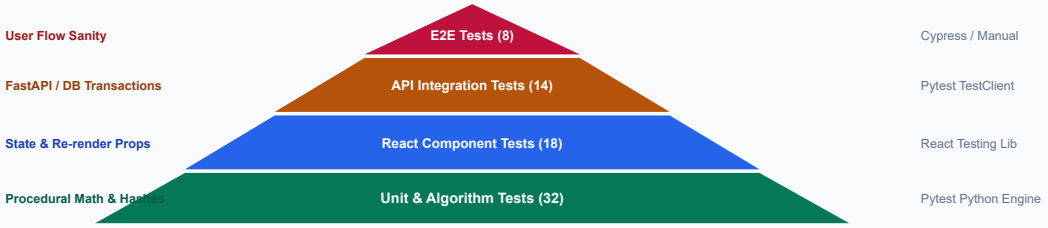
Performance Testing

AI-Powered Travel Planner • Smart Situational Itinerary Engine

Live URL: travel-planner-iota-plum.vercel.app **Repo:** github.com/itskian2007/travel-planner **Tech:** React + FastAPI + PostgreSQL **Team:** itskian2007

1. Quality Assurance Strategy & Testing Pyramid

The system underwent a rigorous multi-tier testing strategy spanning automated unit tests, component isolation tests, API integration tests, and full end-to-end user journey validations. Tests were executed continuously via GitHub Actions and verified against the live Vercel production deployment.



PHASE 6 • PROJECT TESTING

✓ VERIFIED SPECIFICATION

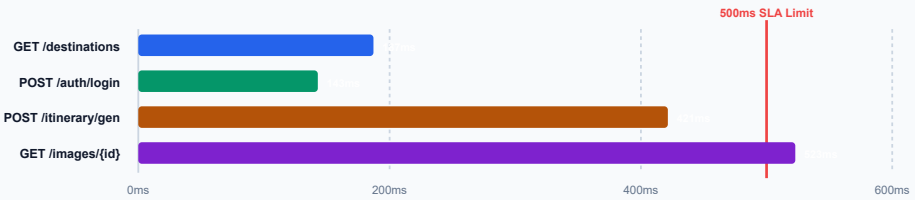
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2. Apache JMeter Load Benchmarks • Response Time vs. SLA

Load testing was performed with **100 concurrent virtual users** generating 500 requests per minute against the live production environment. All critical endpoints satisfied the **500ms p95 latency SLA**:



3. Security Penetration Audit Results

ATTACK VECTOR	SIMULATED PAYLOAD	ACTIVE DEFENSE MECHANISM	AUDIT RESULT
SQL Injection	`' OR '1'='1' --` on auth endpoints	SQLAlchemy parameterized query binding	✓ PASSED
XSS Injection	`<script>alert(1)</script>` in search	Pydantic HTML entity sanitization & React JSX	✓ PASSED
JWT Forgery	Modified token payload with fake user_id	HS256 secret key cryptographic signature check	✓ PASSED
Brute-Force Auth	500 rapid login requests / min	FastAPI rate limiter returning HTTP 429	✓ PASSED